

What Classroom Audio Tells About Teaching: A Cost-effective Approach for Detection of Teaching Practices Using Spectral Audio Features

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Acoustic features and machine learning models have been recently proposed as promising tools to analyze lessons. Furthermore, acoustic patterns, both in the time and spectral domain, have been found to be related to teacher pedagogical practices. Nonetheless, most of previous work relies on expensive or third party equipment, limiting its scalability, and additionally, it is mainly used for diarization. Instead, in this work we present a cost-effective approach to identify teachers' practices according to three categories (*Presenting*, *Administration*, and *Guiding*) which are compiled from the Classroom Observation Protocol for Undergraduate STEM. Particularly, we record teachers' lessons using low-cost microphones connected to their smartphones. We then compute the mean and standard deviation of the amplitude, Mel spectrogram, and Mel Frequency Cepstral coefficients of the recordings to train supervised models for the task of predicting three categories compiled from the Classroom Observation Protocol for Undergraduate STEM. We found that spectral features perform better at the task of predicting teachers' activities along the lessons and that our models can predict the presence of the two most common teaching practices with over 80% of accuracy and good discriminative power. Finally, with these models, we found that using audio obtained from the teachers' smartphones it is also possible to automatically discriminate between sessions where students are using or not an online platform. This approach is important for teachers and other stakeholders who could use an automatic and cost-effective tool for analyzing teaching practices.

CCS Concepts: • **Applied computing** → **Computer-assisted instruction**; • **Information systems** → **Speech / audio search**; • **Human-centered computing** → **Sound-based input / output**; • **Computing methodologies** → **Supervised learning by classification**;

Keywords: Spectral Audio Features, Teacher Activity Detection, Classroom Sound, COPUS

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1 INTRODUCTION

Teacher observation is a crucial tool for both measuring the impact of improvement efforts regarding teaching practices and developing research about the quality of education since it allows to identify particular dimensions in which teaching can be improved that cannot be pointed out by test scores [16]. In fact, it has been proven that by analyzing their own lessons, teachers help to improve their students' academic achievement [30]. Moreover, several teacher education programs are shifting the focus on their content towards high-leverage teaching practices such as eliciting and interpreting students' thinking, leading discussions, and generating positive classroom climate [22]. Nonetheless, many teacher evaluation tools until the last decade offer information about beginning teachers' skills with respect to broad domains of work like planning, instruction, or assessment. This has generated a need for additional tools to improve estimates of teachers' skill regarding particular practices[32]. For this purpose, several instruments that account for these particular practices have been developed. Some of these include the Stallings method [26], the Teaching Dimensions Observation Protocol (TDOP) [12], the Practical Observation Rubric To Assess Active Learning (PORTAAL) [7], the Classroom Assessment Scoring System (CLASS) [29], and the Classroom Observation Protocol for Undergraduate STEM (COPUS) [34]; where observers classify teachers' practices according to different criteria.

In the context of observing STEM (Science Technology Engineering and Mathematics) lessons, the COPUS protocol has been widely used. This tool aims to characterize the general state of teaching, provide feedback to instructors who desired information about how they and their students were spending class time, and identify faculty professional development needs. Even though COPUS was initially thought as a tool to code undergraduate STEM lessons, there is evidence that the COPUS instrument also accounts for important activities that may be present in middle and high school STEM classrooms [1]. COPUS classifies student and teacher actions in 25 different codes, 13 for student actions and 12 for teacher actions, that capture the different ways in which students and their instructors interact along the lessons. One of the advantages of this tool is the minimal amount of training that an observer needs to ensure a reliable classification of teacher and student actions (around 1.5 hours of training). Moreover, COPUS observers only annotate what teachers are explicitly doing without judging the cognitive level of the activities, therefore achieving high values of Inter Rater Reliability [34]. However, this does not exempt the procedure of observing and coding these lectures of being really time consuming, the need of previously trained human observers, and coder bias. To overcome these limitations, different methods have been used to automatically register information about how lessons are developed and identify important variables that can denote the usage of particular teaching practices. In educational research regarding teaching observation, the procedure is usually accompanied with audio or video recordings. However, a large portion of the work in this matter focuses on using linguistic features extracted from Automatic Speech Recognition (ASR) systems and Natural Language Processing (NLP) approaches for modelling teaching practices [19]. This causes these analyses to disregard important information besides the linguistic sources, such as paralinguistic or contextual ones, which are inferable from audio recordings [33], [10].

In the field of audio analytics several systems for classroom activity detection, based on machine learning techniques, have been developed by researchers. Wang et al. [37] used the Learning Environment Analysis (LENA) system [9] and a random forest model to classify elementary school recordings as teacher lecturing, whole-class discussion, and student group work. Donnelly et al. [6] trained seven different machine learning models for the task of predicting question authenticity from teachers' discourse. Owens et al. [27] developed an automated model for decibel analysis using decision trees and identified different teaching patterns related to active learning situations by analyzing the mean volume and its standard deviation along the lessons. James et al. [14] used low-level audio features and conversational features to classify the overall climate of lessons according to the CLASS protocol and revealed that spectral features are better predictors for classroom climate as they carry emotional information because of their dependency on vocal folds.

These works present novel approaches by showing how acoustic features and machine learning methods can be used to measure different teaching dimensions. Nonetheless, most of the tools developed so far rely heavily on particular sensors/devices needed to capture classroom information or register sensitive information from students, which inherently restricts their scalability [37], [6], [27]. Besides, many of them are more focused on diarization tasks (i.e.: identifying who is talking) rather than analyzing what teachers are doing. Thus, to make the idea of using observation protocols feasible at a large scale, there is a need for a tool that can rapidly analyze multiple lessons in short time to give teachers timely feedback [2]. We conceive a scalable and cost-effective tool as one that does not register sensitive information from the students, does not rely on third parties or expensive equipment, and is easy-to-use for teachers in the sense that it requires minimal training and effort to be used effectively. Therefore, given the advances in the matter, we seek to study to which extent teachers' actions along the lesson can be inferred using a non-intrusive, cost-effective, scalable, and automatic approach. This gives rise to our main research question:

- To which extent can we infer important characteristics of teachers' practices from acoustic features using a cost-effective, non-intrusive, and scalable approach?

To answer this question, we studied 42 lessons from 19 teachers in the context of a Randomized Controlled Trial (RCT) that measures the impact of a technology enhanced learning tool, ConectaIdeas, in 4th grade students in Chile [3]. During the study, teachers were observed using the codes related to teaching practices from the COPUS and their speech was recorded using lavalier microphones connected to teachers' smartphones. To achieve our purpose, we first compiled the 12 COPUS codes for teacher observation into four categories: *Presenting*, *Guiding/Feedback*, *Administration*, and *Other*. Secondly, three widely used audio features were computed at a 1-second resolution from the recordings. Next, the recordings and the observations provided by the researchers from [3] are synchronized at the same resolution and four supervised models are trained separately using the previously calculated features as predictors and the synchronized observations as targets. After this, we test the models over a different set of lectures and assess them using precision, recall, accuracy, and the Area Under the Curve (AUC) score. In this way, our approach focuses on giving fast feedback (around 7 minutes of processing for 1 hour of recording) with them only needing to wear the lavalier microphones and connect them to their smartphones. Additionally, our models do not register sensitive information neither from students nor from teachers. Finally, we explore the potential of these models to distinguish between treatment and control groups by analyzing how teachers distributed time amongst the different COPUS categories. We expect this approach to be useful for different educational stakeholders, including teachers and instructors, but also institutions and government agencies.

We give a brief review on relevant previous work in Section 2. Section 3 presents the description of the methodology used to extract the audio features, train the models, and assess them. Section 4 shows the results for the benchmarking of the different models, some examples on the prediction curves versus the ground truth coding, and an exploratory analysis regarding the relation between treatment and control groups in the intervention involving the data and the predictions of the models. Finally, in Section 5 we discuss the methodological implications, contributions, limitations, and possible future work in this matter.

2 RELATED WORK

The usage of machine learning techniques to automate the analysis of classroom discourse from audio has increased in the last decade and promising results have been reported. For example, Wang et al. [37] asked 12 elementary teachers to wear the LENA system while they taught their regular mathematics lessons. Using the LENA system, they calculated for every 30-second window in the lessons the percentage of teacher talk time, the percentage of student talk time, the percentage of overlapping talking time, the percentage of silence time, and the average amplitude. The windows were also annotated by two independent human coders as being primarily “teacher lecturing”, “whole class discussion” or “student group work”. Finally, using the conversational features extracted from the LENA system they trained a random forest model to predict for the dominant activity in each window with results that converged well with the human-coded classification.

Donnelly et at. [6] collected recordings from 11 teachers across 37 lessons and used voice activity detection techniques to extract teachers’ speech, filter the natural noise of middle schools, and segment the recordings in utterances. The researchers also manually coded the 10080 utterances as containing or not containing a question. Later, they extracted a total of 218 linguistic, acoustic, and context features using three automatic speech recognition (ASR) systems, the OpenSmile tool [8], and variable selection methods based on the multicollinearity between the features. Finally, they trained 6 machine learning classifiers using implementations from the WEKA toolkit [38]: Naive Bayes, logistic regression, random forest, J48 decision tree, J48 with Bagging, Bayesian network, and k-nearest neighbors obtaining weighted F-scores of 0.69 for the task of classifying question authenticity.

Owens et al. [27] collected over 1700 hours of audio from more than 60 different courses and developed the Decibel Analysis for Research in Teaching (DART) algorithm which has been the state-of-the-art baseline since then. This model consists of an ensemble of binary decision trees trained on 45 audio recordings from 8 instructors. The recordings were classified and timestamped by three independent coders as single voice, multiple voice, no voice, and other. Subsequently, for each recording the mean and standard deviation of the volume was calculated on a 15-second window at a 0.5-second rate. Next, the model was trained through stratified cross validation [17] and tested over a new set consisting of one recording for each course annotated independently by two new human coders obtaining 90% and 87% of accuracy in the training and test set, respectively. Finally, the researchers were able to identify active learning situations in these courses by recognizing patterns in the volume curves of the recordings. Furthermore, they also analyzed to which extent these situations occur in different undergraduate and major courses with interesting qualitative results.

James et al. [14] collected data from 92 preschool classrooms and classified the overall climate of the lessons as positive or negative classroom climate according to the CLASS protocol. Afterward, they removed the silence from the recordings via thresholding and performed speaker diarization using the LIUM [25] toolkit. From the segmented audio, they extracted conversational features (e.g.: normalized speaker time, speaker duration) from the LIUM clusters and also computed low-level audio features (e.g.: Mel Frequency Cepstral Coefficients (MFCCs) [20], pitch) for each segment using OpenSmile. Finally, Kruskal-Wallis test and correlation based algorithms were used for variable selection and nine different machine learning models were trained using 10-fold cross validation. The models presented accuracy values between 70-80% at the task of predicting classroom climate. The authors also found during the variable selection procedure that spectral features were the most correlated to the outcome variable as they usually carry emotional information due to their dependency on the tension of vocal folds as shown by [10], [33].

These works have shown the potential of machine learning algorithms to classify and analyze classroom audio. Moreover, they have shown that the outputs of these models are closely related to different teaching strategies across multiple educational levels and have identified patterns present in the extracted features that correspond to these strategies. Nonetheless, most of these tools were developed using expensive/third party equipment which limits the possibility of using them to analyze a large population of teachers. Besides, these works focus mainly on the task of recognizing whether the sound comes from teacher speech, student speech, overlapping voices, silence, or noise and then manually associates patterns in these predictions to different teaching practices. This means these algorithms can provide information about who is talking, but not directly about what those people are doing. Instead, considering the performance of the current state-of-the-art tools for classroom audio analysis, we seek to study the possibility of a deeper analysis that involves an automatic classification of teaching practices in three categories (*Presenting*, *Guiding/Feedback*, and *Administration*) from the recordings of their lessons using a cost-effective and non-intrusive method to maximize scalability.

3 METHODOLOGY

3.1 Dataset

The raw dataset consisted of 47 audio recordings from an RCT carried out to measure the impact of using a technology guided platform, ConectaIdeas, in fourth graders from low-performing schools in Chile. The platform was implemented using exercises from the “Plan de Apoyo Compartido” (PAC) program, conducted by the Ministry of Education during 2011 and 2012 [5]. During the RCT, the treatment group performed weekly lab sessions with ConectaIdeas, while the control group performed their usual lessons. Student performance was evaluated using a calibrated test developed by MIDE UC, Pontificia Universidad Católica de Chile, which has been validated with a Cronbach's alpha of 0.91 [21]. The implementation involved a total of 19 teachers, where number of recordings per teacher varied from one to 11, while the length of these ranged from 18 to 77 minutes. The audio was recorded using Rode SmartLav+ microphones connected to teachers’ smartphones at 44100 Hz.

Additionally, for each lesson in our dataset, a trained observer used a smartphone application [18] to classify teachers’ practices in real-time by pressing and releasing buttons that corresponded to the COPUS codes. These buttons can be pressed simultaneously and independently. This means both, the original observations and the compiled categories might intersect along time. Each teacher code of the COPUS protocol was tagged with the timestamps for the beginning and the end, both with respect to the start of the observation, and the ID of the corresponding recording. Later, the original 12 COPUS codes related to teaching practices were compiled into four categories: *Presenting*, *Guiding/Feedback*, *Administration*, and *Other* to ease analysis and visualization, as they naturally group this way according to [34]. Table 1 presents the mapping from COPUS codes to our categories, as well as a brief description on the meaning of the codes according to [34]. Lastly, audio and the COPUS observations were needed to train and test the models. Thus, the dataset used to train the models consisted of 42 recordings with their respective COPUS synchronization (34 hours of audio).

Table 1: Categories used to train the classification models. Each category presents the codes that compose it and a brief description of each code.

Code	Code Description
<i>Presenting</i>	
Lec	Lecturing (Presenting content, deriving mathematical results, presenting a problem, solution, etc.)
RtW	Real-time writing on board, doc, projector, etc. (often checked along with Lec)
D/V	Showing or conducting a demo, experiment, simulation, video, or animation
<i>Guiding</i>	
FUp	Follow-up/feedback on clicker question or activity to entire class
PQ	Posing non-clicker question to students
CQ	Asking a clicker question (entire time using the clicker question, not just when first asked)
AnQ	Listening to and answering student questions with entire class listening
MG	Moving through class guiding ongoing student work during active learning task
101	One-on-one extended discussion with one or few individuals, not paying attention to the rest of the class.
<i>Administration</i>	
Adm	Administration (assign homework, return tests, etc.)
<i>Other</i>	
W	Waiting when there is an opportunity for an instructor to be interacting with or observing/listening to student or group activities while the instructor is not doing so.
O	Other – Explain in comments

3.2 Feature Engineering

To prepare the recordings for the models, we divided each one into 1-second segments (122448 samples in total). For each segment we calculated the mean and the standard deviation of the amplitude in windows that ranged from five to 17 seconds centered at such segment. Later, we calculated two spectral features over these windows

which have been shown to achieve state-of-the-art performance in different tasks regarding audio processing [13]: a decibel-scaled Mel spectrogram and the first 13 components of the MFCCs.

The procedure to extract the Mel spectrogram from an audio signal involves several steps. First of all, the signal is divided into small windows (around 20ms). Second, the Short Time Fourier Transform (STFT) is computed for each window, transforming them from the time to the frequency domain. Third, the frequency axis is re-scaled using the Mel scale and partitioned into N (64) equally spaced bands. The Mel scale is a logarithmic transformation of an affine function with respect to the frequency axis in Hertz which is based on the fact that humans perceive differences in lower frequencies better than in higher ones. Therefore, the equally spaced bands in the Mel scale should “sound equally spaced” to the listener. Fourth, once the STFTs of each window are calculated and mapped to the N bands, they are stacked along the time axis and the amplitude is mapped to the decibel scale to obtain the spectrogram.

Subsequently, in the same way we did with the amplitude, we calculated the mean and the standard deviation along each decibel-scaled Mel band. Aside from the Mel spectrogram, we also calculated the mean and standard deviation along the first 13 coefficients from the MFCCs. These coefficients are obtained by applying the discrete cosine transform to the logarithms of the powers in a Mel-scaled power spectrum of the signal. The spectral features were computed using librosa library for python [23]. Next, each feature was normalized with respect to the whole lesson to lessen the influence of background noise, giving a total of 156 audio features for each category and window length combination. Figure 1 presents two examples of Mel Spectrograms calculated from two intervals where teachers were coded as being presenting content and two from non-*Presenting* intervals.

Once the audio processing was finished, each segment was tagged with boolean labels corresponding to the three COPUS categories by synchronizing the timestamps of the observations and the recordings at a 1s resolution. The segments that presented none of the categories were not considered for the models as they could not be trusted as true negatives. Thus, the final dataset consisted of 115402 samples with 160 features each. It must be noticed that the compiled categories are not disjoint along the lessons. Therefore, a binary model was trained to predict for each one of them. Finally, two teachers who did not have lessons in the training set, were of different genders, and belonged to different groups during the RTC were selected and their respective lessons were used as a test set while the rest was left as the training set. This way, we can estimate the performance of our models in unseen data. Table 2 presents the proportions of the ground truth COPUS categories across the 42 lessons. Due to the lack of representation of Other in the dataset and because the 10 (out of the 12) COPUS codes that relate to particular teaching practices were covered by the other three categories, we decided to drop this category from the following analyses.

Table 2: Percentage of positive samples for the ground truth categories across all lessons. Please notice the categories might intersect along time, therefore, this does not represent a probability distribution.

Category	Positive Samples	Percentage
Presenting	33209	28.8%
Guiding	71117	61.6%
Administration	11079	9.6%
Other	2347	2.0%
Total	115402	

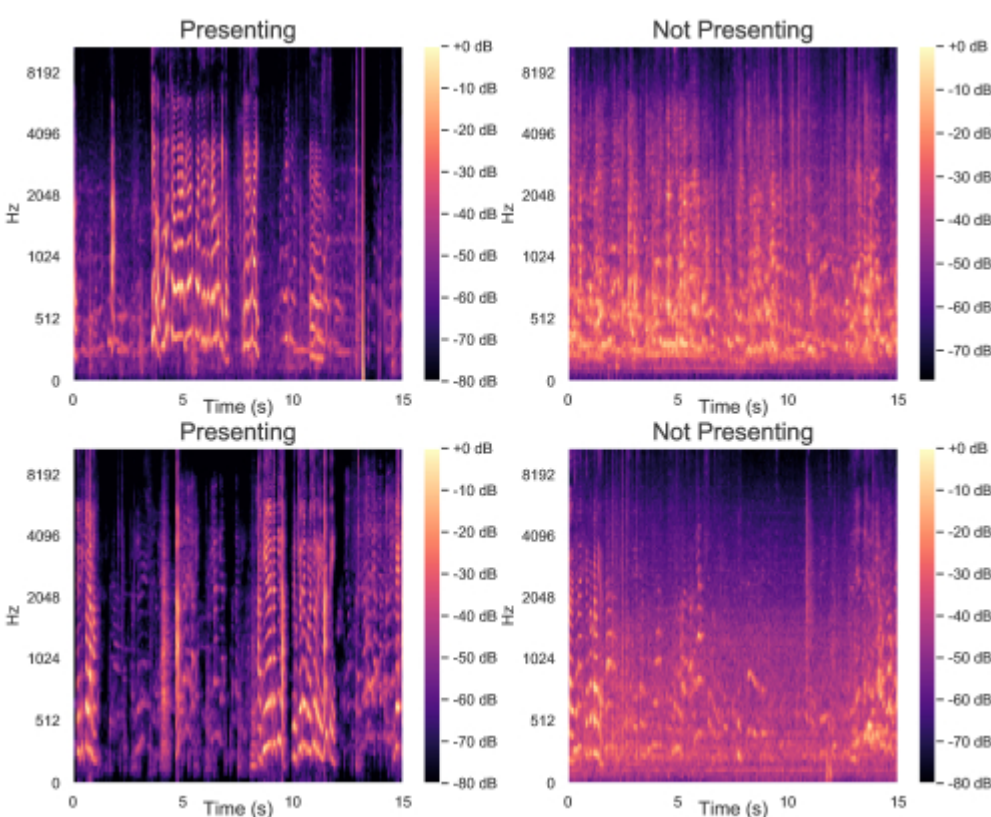


Figure 1: decibel-scaled Mel spectrograms for 15-second windows extracted from the lessons. Spectrograms at the left are extracted from windows classified as *Presenting* and the ones at the right are extracted from non-*Presenting* intervals.

3.3 Model Training and Evaluation

3.3.1 Random Forest. Random forests [4] were used to predict the presence of COPUS categories using the acoustic features. These models consist of an ensemble of (in this case binary) decision trees [31]. In a random forest, each decision tree is fitted with a bootstrapped dataset extracted from the original and within each node of such tree, the feature that best separates the data contained in the node according to a certain criterion (usually Gini impurity or entropy) is selected from a random subset of the whole variable set. Then the data is split according to that feature and two child nodes are created with the partitioned data. The nodes that have no children are called leaf nodes of the tree. The tree starts with one node containing all the data and the procedure of splitting leaf nodes is repeated until the tree reaches a predefined maximum depth, every leaf node reaches a predefined threshold for the split criterion, or until the data in each leaf node belongs to the same class. What mainly differentiates a random forest from a decision tree is that both of the techniques stated above (bootstrapping and feature sampling) have been shown to reduce the common overfitting problems that appear in tree-based models and enhance their capability to predict over unseen data despite their counter-intuitive nature.

3.3.2 Parameter Tuning and Model Evaluation. To estimate suitable values for the window length used to extract the audio features and the maximum depth of the trees in the random forest models a five-fold stratified cross-validation [17] was performed over the training set for each category and feature combination. Stratified cross-validation is a brute-force search method used to estimate the optimal parameters for a model. When performing cross validation, the data is partitioned into K (in this case five) disjoint folds. A partition is said to be stratified when those folds are chosen so that each one has roughly the same proportion of each target class, therefore acquainting for how represented the classes in the whole training data are. Next, for each parameter combination the model is repeatedly trained over K-1 folds and tested over the remaining one, and performance measures (in this case accuracy) are calculated over the testing fold. Finally, the combination of parameters that yielded the best average accuracy across folds is chosen for the model. In this case, the parameters considered for the cross validation were the length of the window centered at each sample (5, 7, 9, 11, 13, 15 and 17s) and the maximum depth of the decision trees that composed such random forests. Table 3 summarizes the best parameters obtained for each COPUS category and feature combination over the training set.

Table 3: Optimal values for window length and maximum depth of the random forest obtained by 5-fold stratified cross-validation for the different category-feature combinations. AMPL, MEL, and MFCC stand for the amplitude, Mel spectrogram, and MFCCs as features respectively.

Category	Feature	Window Length	Maximum Depth
Presenting	AMPL	7s	4
	MEL	11s	5
	MFCC	17s	6
Guiding	AMPL	9s	5
	MEL	17s	4
	MFCC	17s	6
Administration	AMPL	11s	2
	MEL	15s	4
	MFCC	15s	4

Once the optimal parameters were chosen for each feature and category combination, the models were retrained over the whole training set. Next, three classification assessment metrics [36] were calculated (precision, recall and accuracy) and a ROC curve analysis [11] was performed over both the training and test set for each model to determine optimal cutoff values for a prediction to be considered positive and to compare their capabilities of predicting COPUS categories. The models were trained and tested using scikit-learn library [28]. Figure 2 summarizes the whole procedure for the feature engineering and model evaluation and training.

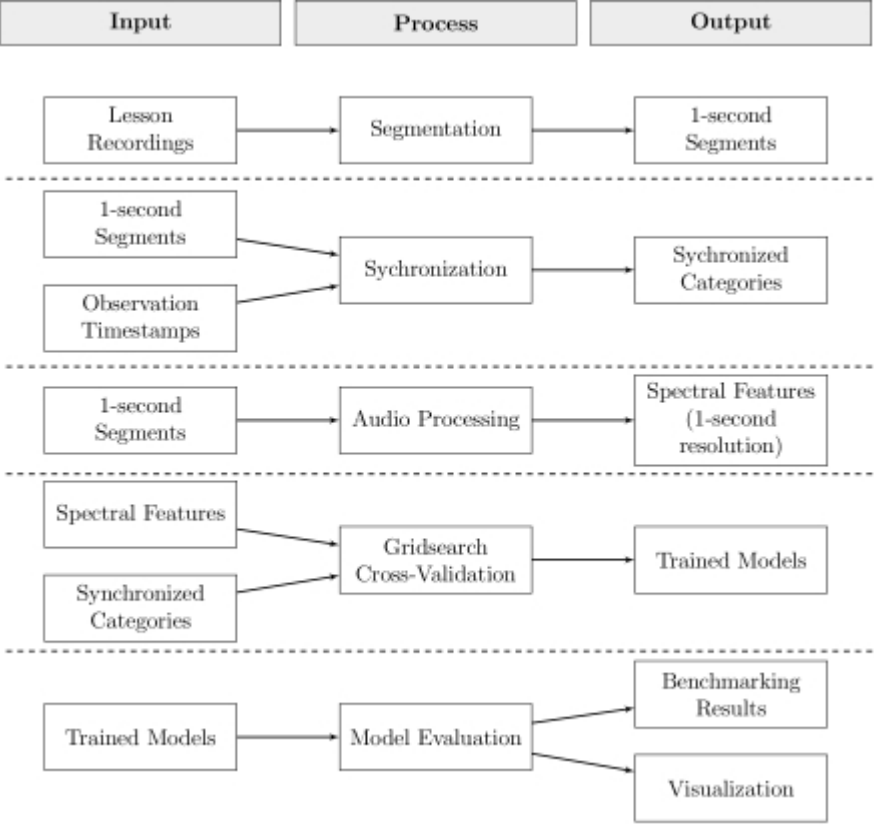


Figure 2: Pipeline for the feature engineering, synchronization, training and evaluation procedures to develop a classifier for COPUS categories.

4 RESULTS

4.1 Benchmarking for the Different Models

To evaluate the quality of the models, three classification assessment metrics were calculated, these are: accuracy, precision and recall. Accuracy represents the percentage of instances that were correctly classified by the model (either positive or negative), precision measures the ratio between the true positives and the instances classified as positive by the model and (1- type I error), and recall measures the ratio of originally positive samples that were classified as positive by the model (1- type II error). Aside from these three metrics, an AUC analysis was performed to find suitable thresholds for the predictions to be considered positive by the models and to examine how the true and false positive rates varied depending on such threshold. We considered the random forest model with the mean and standard deviation of the amplitude as our baseline, since these features have been proven to be suitable for classroom audio analysis by [37]. We then compared the other three models against the baseline.

The results show that spectral features generally perform better than the amplitude, increasing both the amount of correctly classified instances (accuracy) and the discriminative power of the model (AUC/ROC score). Table 4 summarizes the results of the different models for *Presenting*, *Guiding/Feedback* and *Administration*. Additionally, the predictions obtained by the best models converge well with human annotations, particularly in the two most represented categories (*Presenting* and *Guiding/Feedback*), as the curves along time are visually able to recognize prolonged periods of these practices which is the most common way they appear across the lessons (see Figure 3).

Table 4: Precision, Recall, Accuracy and AUC over the test set for models using optimal parameters obtained through cross validation. Best feature sets by category in bold.

Model	Precision (o/1)	Recall (o/1)	Accuracy	AUC/ ROC
<i>Presenting</i>				
AMPL (7s)	0.82 / 0.91	0.99 / 0.09	0.82	0.81
MEL (11s)	0.84 / 0.87	0.99 / 0.22	0.84	0.84
MFCC (17s)	0.86 / 0.83	0.98 / 0.35	0.86	0.87
AMPL+MEL +MFCC	0.85 / 0.88	0.99 / 0.31	0.86	0.87
<i>Guiding</i>				
AMPL (9s)	0.84 / 0.80	0.21 / 0.99	0.81	0.76
MEL (17s)	0.81 / 0.82	0.32 / 0.98	0.83	0.86
MFCC (17s)	0.72 / 0.82	0.33 / 0.96	0.80	0.84
AMPL+MEL +MFCC	0.75 / 0.84	0.42 / 0.96	0.83	0.85
<i>Administration</i>				
AMPL (11s)	0.96 / 0.00	1.00 / 0.00	0.96	0.46
MEL (15s)	0.96 / 0.19	0.99 / 0.04	0.96	0.62
MFCC (15s)	0.96 / 0.00	1.00 / 0.00	0.96	0.64
AMPL+MEL +MFCC	0.96 / 0.12	0.99 / 0.02	0.96	0.64

However, we found some limitations during model evaluation. First of all, fast and consecutive changes between categories are not always identified, such as the short *Administration* window at the end of the lesson shown in Figure 3. This might be due to the fact that our model averages spectral features over time, possibly mitigating the effect of those fast changes. Secondly, due to the nature of the acoustic features, we found that *Guiding* segments were mainly associated with really high or low levels of classroom noise and a uniform fading in intensity from the lower to the higher bands. On the other hand, *Presenting* samples were usually related to moderated levels of classroom noise with specific bands highlighted due to teachers’ voice and low values in the higher bands, as in Figure 1. This caused *Presenting* segments which had unusually high levels of noise to be misclassified as negative samples while also triggering false positives in the *Guiding* models, as shown in the beginning of Figures 3a and 3b . Finally, poorly represented categories in our data, such as *Administration* and *Other* (9.6% and 2% of the dataset, respectively) struggle to be correctly identified by our models, as the presence of these categories is systematically predicted negative across all feature sets (Figure 3c). In spite of these limitations, we find these results relevant as the models are able to distinguish between the two most common teachers’ actions, which gather 88% of the total lesson time, with high accuracy and discriminative power.

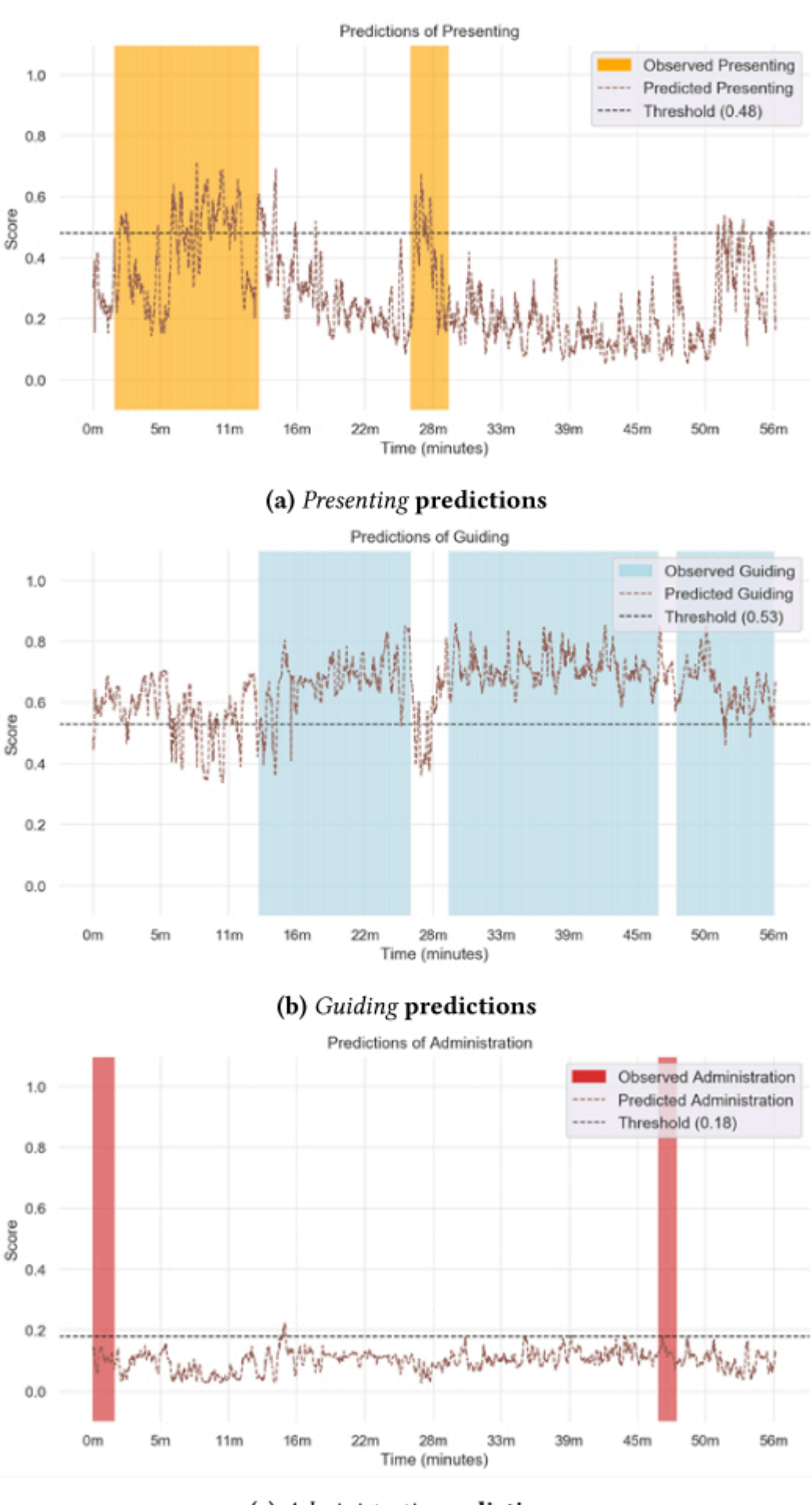


Figure 3: Predictions curves obtained by the best models over a lesson with varied activities. Colored background indicates positive ground truth observation, horizontal line indicates threshold for predictions to be considered positive.

4.2 A Use Case Scenario

As stated before, the dataset was extracted from lessons in the context of an RCT that measured the impact of using a technology guided platform with 4th graders. Therefore, given the particular nature of the data, we were intrigued to explore possible relations between how teachers distributed time in the different categories over their lessons and whether those lessons were developed using ConectaIdeas or carried out as usual. To validate this hypothesis, we started by calculating the proportion of time each teacher spent on the different categories using the ground truth labels and performing a univariate profiling of these quantities (see Figure 4). By doing this, we

found significant differences in the way teachers manage time during their lessons due to the presence of the technology guided platform.

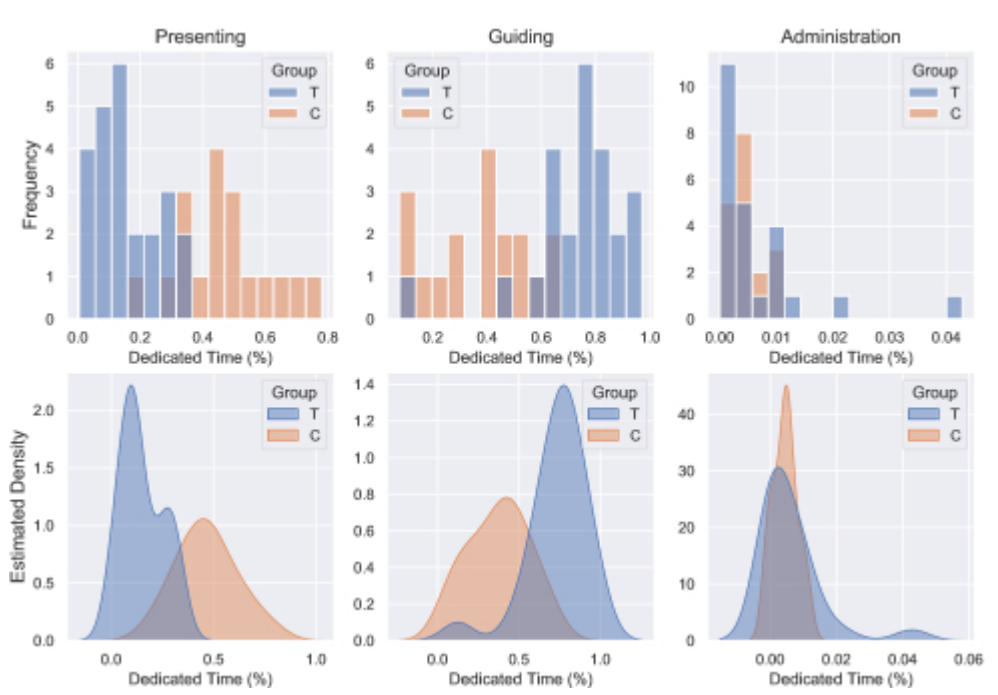


Figure 4: Histogram and kernel density estimations for the proportion of time spent in different COPUS categories across lessons by control and treatment groups, ground truth values. The p-values for the Kruskal-Wallis test from left to right are $3.63\text{e-}7$ for *Presenting*, $4.23\text{e-}7$ for *Guiding/Feedback*, and 0.047 for *Administration*.

The treatment group spent, on average, substantially more time guiding or giving their students feedback than the control group would as teachers from the first dedicated 78% of their lessons to these tasks. On the other hand, teachers in the control group only spent 42% of their time in this category. Additionally, the lessons from the control group were more lecture-based than the ones from the treatment group, spending 44% and 15% of the time presenting content to the students in average, respectively. Lastly, teachers from the control group spent more time in administrative tasks than the treatment group, with averages of 13% and 6%, respectively. Furthermore, Kruskal-Wallis test [24] was performed to ensure the statistical significance of time distribution against the control and treatment groups. We found significant results for the three categories considered in the analysis scoring p-values < 0.05 . These analyses were later repeated using the predictions from the best model for each category obtaining similar distributions to the ones presented by the ground truth values and significant p-values on the Kruskal-Wallis test for the predictions of *Presenting* and *Guiding/Feedback* (Figure 5).

These differences showed that it is possible to characterize treatment and control groups based on their time distributions. Therefore, we wanted to explore the possibility of doing this using the predictions from the models instead of the ground truth coding. For this purpose a simple experiment was performed: we trained two logistic regressions (one with the ground truth values and the other one with the predictions from the models) on the task of predicting whether the lesson was performed under the control or treatment group given the proportion of time spent across the different categories. The models were trained using five-fold cross validation and achieved results of 90% for the ground truth categories and 84% for the predicted categories in terms of accuracy, indicating that automatic predictions from the models using acoustic features can distinguish between two groups implementing different teaching practices due to an external intervention with results that align with the ones obtained using human coding.

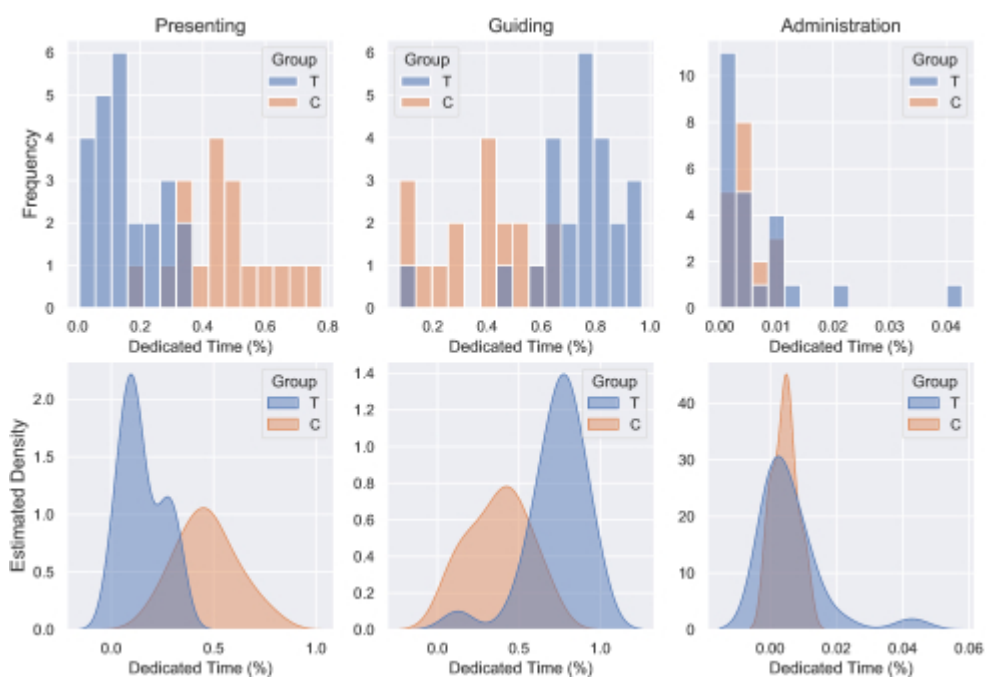


Figure 5: Histogram and kernel density estimations for the proportion of time spent in different COPUS categories across lessons by control and treatment groups, predicted values. The p-values for the Kruskal-Wallis test from left to right are $2.52\text{e-}4$ for *Presenting*, $2.92\text{e-}6$ for *Guiding/Feedback*, and 0.079 for *Administration*.

5 DISCUSSION

In summary, we presented the development and evaluation of a machine learning model that captures different teaching practices considered in the COPUS instrument from the recordings of the lessons using spectral audio features. Furthermore, we showed an application of this tool by analyzing to which extent different teaching practices, ascribable to an intervention using a technology guided platform, can be distinguished using our model. We consider this work sets forth two key contributions over previous research. First of all, our approach is non-intrusive for students as teachers only need to wear lavalier microphones connected to their phones. Besides, this setup ensures that mainly teacher audio is used to train and validate the models, as our central aim is to give teachers timely feedback for them to analyze their time management amongst different practices. Moreover, it does not rely on third party software or expensive equipment other than teachers’ smartphones and budget lavalier microphones. Consequently, our approach maximizes cost-effectiveness and scalability when analyzing teaching practices. Secondly, previous research that considers audio features as inputs either approach classroom sound analysis as a diarization task or classify different dimensions of teaching in accordance with other protocols whose aim differs from ours. Additionally, since we used cross-validation to train the models and tested over a separate set of lessons, we avoid biasing them on particular teachers’ behaviors. Moreover, since we use spectral audio features, which are usually associated with paralinguistic features rather linguistic ones, we expect the models to be robust when varying the content of the lessons as our approach should not depend on particular keywords. However, these are still open questions and these hypotheses should be tested via future research.

Results show that our models can differentiate between lecturing and guiding periods and are able to estimate the presence of these actions along the lesson with relatively high accuracy at 86% and 83% and good discriminative power at 0.87 and 0.86 of AUC/ROC score for *Presenting* and *Guiding*, respectively. Moreover, the predictions show similar patterns with respect to human annotations and seem to be sufficient to discern between the two types of lessons present in our dataset. However, our models are not exempt of limitations. First, that they systematically reject the presence of the least represented categories in our dataset (*Administration* and *Other*). This is a common issue amongst machine learning classifiers when working with considerable class imbalance [15]. Thus, we expect these categories can be better captured by training the models over a more balanced dataset. Secondly, high levels of classroom noise may cause false positives/negatives when predicting *Guiding/Feedback* and *Presenting* respectively, since segments from the latter were generally characterized by a clear distinction of teacher's voice. We believe this can be overcome by complementing our analyses with other features such as linguistic ones, which are able to distinguish between speech and other audio sources.

Finally, we propose three different directions for future research that emerge naturally from the results and limitations found during our work. First, given that recent research has shown the utility of linguistic keywords for other tasks regarding automatic content analysis, we propose complementing the analysis of the spectral audio features with a NLP approach to build a multimodal learning model. In this way it would be possible to account for paralinguistic and linguistic features when inferring teachers’ actions. Secondly, the predictions from the models converged well with the block structure we observed over the ground truth coding of the lessons along time. However, since our models were conceived to predict at a one second resolution, there are short windows contained in these blocks that are not correctly classified. Therefore, given this structure, it might be possible to extract the different “blocks” from a lesson using a post-processing over the raw predictions that imposes this structure. Third, since there exist already standardized algorithms for computing spectral features on mobile phones [35], the procedure for recording, processing and analyzing the recordings could be completely implemented on the teachers’ smartphones. This is particularly important for teachers with unstable or low internet connection, which is a common case upon low Socio-Economic Status schools. Additionally, these models could be used by stakeholders besides individual teachers to characterize teaching practices. However, since our central aim is to give teachers feedback, these models should always be used protecting teachers’ and students’ sensitive data, for example, by aggregating it to higher levels.

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